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Methods and systems for estimating neural activation by stimulation using a stimulation system

Abstract

A method for estimating neural activation arising from stimulation by a stimulation system includes identifying different neural elements stimulated by the stimulation; obtaining a neural response signal resulting from the stimulation by the stimulation system; and decomposing the neural response signal to estimate neural activation of each of the different neural elements.

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11185697	12/2020	Sinclair	N/A	A61B 5/6868
11278726	12/2021	Sinclair	N/A	A61N 1/36167
11285329	12/2021	Carcieri et al.	N/A	N/A
11298550	12/2021	Howard et al.	N/A	N/A
11357986	12/2021	Steinke et al.	N/A	N/A
11517755	12/2021	Zhang et al.	N/A	N/A
11707622	12/2022	Juarez Paz et al.	N/A	N/A
11745010	12/2022	Donega et al.	N/A	N/A
11890478	12/2023	Sinclair	N/A	A61B 5/377
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2004/0181262	12/2003	Bauhahn	N/A	N/A
2004/0186532	12/2003	Tadlock	N/A	N/A
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Patent Application Ser. No. 63/210,799, filed Jun. 15, 2021, which is incorporated herein by reference.

FIELD

(1) The present disclosure is directed to the area of implantable electrical stimulation systems and methods of making and using the systems. The present disclosure is also directed to methods and systems for estimating neural activation by stimulation using the implantable electrical stimulation system.

BACKGROUND

(2) Implantable electrical stimulation systems have proven therapeutic in a variety of diseases and disorders. For example, spinal cord stimulation systems have been used as a therapeutic modality for the treatment of chronic pain syndromes. Peripheral nerve stimulation has been used to treat chronic pain syndrome and incontinence, with a number of other applications under investigation. Deep brain stimulation can be used to treat a variety of diseases and disorders.

(3) Stimulators have been developed to provide therapy for a variety of treatments. A stimulator can include a control module (with a pulse generator) and one or more stimulator electrodes. The one or more stimulator electrodes can be disposed along one or more leads, or along the control module, or both. The stimulator electrodes are in contact with or near the nerves, muscles, or other tissue to be stimulated. The pulse generator in the control module generates electrical pulses that are delivered by the electrodes to body tissue.

BRIEF SUMMARY

(4) One aspect is a method for estimating neural activation arising from stimulation by a stimulation system. The method includes identifying a plurality of different neural elements stimulated by the stimulation; obtaining a neural response signal resulting from the stimulation by the stimulation system; and decomposing the neural response signal to estimate neural activation of

each of the different neural elements.

(5) In at least some aspects, the method further includes, for each of the different neural elements, providing at least one template signal for the neural element, wherein decomposing the neural response signal includes decomposing the neural response signal using the template signals. In at least some aspects, decomposing the neural response signal using the template signals includes determining weights for each of the template signals as portions of the neural response signal.

(6) In at least some aspects, decomposing the neural response signal includes decomposing the neural response signal in a frequency domain. In at least some aspects, the method further includes transforming the neural response signal into the frequency domain. In at least some aspects, the method further includes determining initial weights for each of the template signals using one or more frequency components in the template signals. In at least some aspects, the method further includes combining the template signals using the weights to generate a composite neural signal. In at least some aspects, the method further includes determining a difference between the neural response signal and the composite neural signal. In at least some aspects, the method further includes determining whether the difference exceeds a predetermined threshold difference and, when so, adjusting the weights.

(7) In at least some aspects, decomposing the neural response signal includes decomposing the neural response signal in a time domain. In at least some aspects, the method further includes combining the template signals using the weights to generate a composite neural signal. In at least some aspects, determining a difference between the neural response signal and the composite neural signal. In at least some aspects, the method further includes determining whether the difference exceeds a predetermined threshold difference and, when so, adjusting the weights. In at least some aspects, the method further includes filtering the neural response signal.

(8) Another aspect is a system for estimating neural activation arising from stimulation by a stimulation system. The system includes at least one processor configured to perform actions, including: receiving an identification of a plurality of different neural elements stimulated by the stimulation; receiving a neural response signal resulting from the stimulation by the stimulation system; and decomposing the neural response signal to estimate neural activation of each of the different neural elements. The actions further include any of the methods described above.

(9) In at least some aspects, the actions further include, for each of the different neural elements, obtaining at least one template signal for the neural element, wherein decomposing the neural response signal includes decomposing the neural response signal using the template signals. In at least some aspects, the system further includes a display and at least one input device coupled to the at least one processor, wherein the actions further include, for each of the different neural elements, receive a selection of a type of signal to be used for the template signal.

(10) In at least some aspects, the system further includes a display and at least one input device coupled to the at least one processor, wherein receiving the identification includes displaying a representation of one or more anatomical regions; receiving a selection by the user of one of the anatomical regions; and identifying one or more of the different neural elements based on the selection of the one of the anatomical regions.

(11) A further aspect is a non-transitory computer-readable medium having processor-executable instructions for estimating neural activation arising from stimulation by a stimulation system, the processor-executable instructions when installed onto a device enable the device to perform actions, the actions including: receiving an identification of a plurality of different neural elements stimulated by the stimulation; receiving a neural response signal resulting from the stimulation by the stimulation system; and decomposing the neural response signal to estimate neural activation of each of the different neural elements. The actions further include any of the methods described above.

(12) In at least some aspects, the actions further include, for each of the different neural elements,

obtaining at least one template signal for the neural element, wherein decomposing the neural response signal includes decomposing the neural response signal using the template signals.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Non-limiting and non-exhaustive embodiments of the present invention are described with reference to the following drawings. In the drawings, like reference numerals refer to like parts throughout the various figures unless otherwise specified.
- (2) For a better understanding of the present invention, reference will be made to the following Detailed Description, which is to be read in association with the accompanying drawings, wherein:
- (3) FIG. 1 is a schematic view of one embodiment of an electrical stimulation system;
- (4) FIG. 2A is a schematic illustration of one embodiment of a neural response or neural signal;
- (5) FIG. 2B is a schematic illustration of another embodiment of a neural response or neural signal;
- (6) FIG. 3 is a flowchart of one embodiment of a method for estimating neural activation arising from stimulation by a stimulation system;
- (7) FIG. 4 is a flowchart of one embodiment of a method for decomposing the recorded neural response signal in the frequency domain;
- (8) FIG. 5 is a flowchart of one embodiment of a method for decomposing the recorded neural response signal in the time domain;
- (9) FIG. 6 is a schematic representation of one embodiment of a user interface for selection of anatomical regions;
- (10) FIG. 7A is a schematic representation of one embodiment of a user interface for selecting which neural signals to use for the neural elements;
- (11) FIG. 7B is a schematic representation of one embodiment of three template signals for neural elements;
- (12) FIG. 7C is a schematic representation of one embodiment of a composite signal using a weighted combination of the three template signals of FIG. 7B;
- (13) FIG. 7D is a schematic representation of one embodiment of a user interface for selecting a weighting strategy;
- (14) FIG. 8 is a schematic representation of neural signal before and after filtering;
- (15) FIG. 9 is a schematic representation of one embodiment of a user interface for selecting filters or other signal processing;
- (16) FIG. 10 is a schematic representation of one embodiment of a user interface for selecting filters or other signal processing;
- (17) FIG. 11 is a schematic representation of one embodiment of a user interface for inputting stimulation parameters;
- (18) FIG. 12 is a schematic representation of one embodiment of determined weights for a stimulation;
- (19) FIG. 13 is a schematic overview of one embodiment of components of a stimulation system, including an electronic subassembly disposed within a control module; and
- (20) FIG. 14 is a schematic block diagram of a system for practicing the methods described herein.

DETAILED DESCRIPTION

(21) The present disclosure is directed to the area of implantable electrical stimulation systems and methods of making and using the systems. The present disclosure is also directed to methods and systems for estimating neural activation by stimulation using the implantable electrical stimulation system.

(22) Suitable implantable electrical stimulation systems include, but are not limited to, a least one lead with one or more electrodes disposed on a distal portion of the lead and one or more terminals

disposed on one or more proximal portions of the lead. Leads include, for example, percutaneous leads, paddle leads, cuff leads, or any other arrangement of electrodes on a lead. Examples of electrical stimulation systems with leads are found in, for example, U.S. Pat. Nos. 6,181,969; 6,516,227; 6,609,029; 6,609,032; 6,741,892; 7,244,150; 7,450,997; 7,672,734; 7,761,165; 7,783,359; 7,792,590; 7,809,446; 7,949,395; 7,974,706; 8,175,710; 8,224,450; 8,271,094; 8,295,944; 8,364,278; 8,391,985; and 8,688,235; and U.S. Patent Application Publication Nos. 2007/0150036; 2009/0187222; 2009/0276021; 2010/0076535; 2010/0268298; 2011/0005069; 2011/0004267; 2011/0078900; 2011/0130817; 2011/0130818; 2011/0238129; 2011/0313500; 2012/0016378; 2012/0046710; 2012/0071949; 2012/0165911; 2012/0197375; 2012/0203316; 2012/0203320; 2012/0203321; 2012/0316615; 2013/0105071; and 2013/0197602, all of which are incorporated herein by reference. In the discussion below, a percutaneous lead will be exemplified, but it will be understood that the methods and systems described herein are also applicable to paddle leads and other leads.

(23) A percutaneous lead for electrical stimulation (for example, deep brain, spinal cord, or peripheral nerve stimulation) includes stimulation electrodes that can be ring electrodes, segmented electrodes that extend only partially around the circumference of the lead, or any other type of electrode, or any combination thereof. The segmented electrodes can be provided in sets of electrodes, with each set having electrodes circumferentially distributed about the lead at a particular longitudinal position. A set of segmented electrodes can include any suitable number of electrodes including, for example, two, three, four, or more electrodes. For illustrative purposes, the systems and leads are described herein relative to use for deep brain stimulation, but it will be understood that any of the leads can be used for applications other than deep brain stimulation, including spinal cord stimulation, peripheral nerve stimulation, dorsal root ganglion stimulation, sacral nerve stimulation, or stimulation of other nerves, muscles, and tissues.

(24) Turning to FIG. 1, one embodiment of an electrical stimulation system **10** includes one or more stimulation leads **12** and an implantable pulse generator (IPG) **14**. The system **10** can also include one or more of an external remote control (RC) **16**, a clinician's programmer (CP) **18**, an external trial stimulator (ETS) **20**, or an external charger **22**. The IPG and ETS are examples of control modules for the electrical stimulation system.

(25) The IPG **14** is physically connected, optionally via one or more lead extensions **24**, to the stimulation lead(s) **12**. Each lead carries multiple electrodes **26** arranged in an array. The IPG **14** includes pulse generation circuitry that delivers electrical stimulation energy in the form of, for example, a pulsed electrical waveform (i.e., a temporal series of electrical pulses) to the electrode array **26** in accordance with a set of stimulation parameters. The implantable pulse generator can be implanted into a patient's body, for example, below the patient's clavicle area or within the patient's buttocks or abdominal cavity or at any other suitable site. The implantable pulse generator can have multiple stimulation channels which may be independently programmable to control the magnitude of the current stimulus from each channel. In some embodiments, the implantable pulse generator can have any suitable number of stimulation channels including, but not limited to, 4, 6, 8, 12, 16, 32, or more stimulation channels. The implantable pulse generator can have one, two, three, four, or more connector ports, for receiving the terminals of the leads and/or lead extensions.

(26) The ETS **20** may also be physically connected, optionally via the percutaneous lead extensions **28** and external cable **30**, to the stimulation leads **12**. The ETS **20**, which may have similar pulse generation circuitry as the IPG **14**, also delivers electrical stimulation energy in the form of, for example, a pulsed electrical waveform to the electrode array **26** in accordance with a set of stimulation parameters. One difference between the ETS **20** and the IPG **14** is that the ETS **20** is often a non-implantable device that is used on a trial basis after the neurostimulation leads **12** have been implanted and prior to implantation of the IPG **14**, to test the responsiveness of the stimulation that is to be provided. Any functions described herein with respect to the IPG **14** can likewise be performed with respect to the ETS **20**.

(27) The RC **16** may be used to telemetrically communicate with or control the IPG **14** or ETS **20** via a uni- or bi-directional wireless communications link **32**. Once the IPG **14** and neurostimulation leads **12** are implanted, the RC **16** may be used to telemetrically communicate with or control the IPG **14** via a uni- or bi-directional communications link **34**. Such communication or control allows the IPG **14** to be turned on or off and to be programmed with different stimulation parameter sets. The IPG **14** may also be operated to modify the programmed stimulation parameters to actively control the characteristics of the electrical stimulation energy output by the IPG **14**. The CP **18** allows a user, such as a clinician, the ability to program stimulation parameters for the IPG **14** and ETS **20** in the operating room and in follow-up sessions. Alternately, or additionally, stimulation parameters can be programed via wireless communications (e.g., Bluetooth) between the RC **16** (or external device such as a hand-held electronic device) and the IPG **14**.

(28) The CP **18** may perform this function by indirectly communicating with the IPG **14** or ETS **20**, through the RC **16**, via a wireless communications link **36**. Alternatively, the CP **18** may directly communicate with the IPG **14** or ETS **20** via a wireless communications link (not shown). The stimulation parameters provided by the CP **18** are also used to program the RC **16**, so that the stimulation parameters can be subsequently modified by operation of the RC **16** in a stand-alone mode (i.e., without the assistance of the CP **18**).

(29) For purposes of brevity, the details of the RC **16**, CP **18**, ETS **20**, and external charger **22** will not be further described herein. Details of exemplary embodiments of these devices are disclosed in U.S. Pat. No. 6,895,280, which is expressly incorporated herein by reference. Other examples of electrical stimulation systems can be found at U.S. Pat. Nos. 6,181,969; 6,516,227; 6,609,029; 6,609,032; 6,741,892; 7,949,395; 7,244,150; 7,672,734; and 7,761,165; 7,974,706; 8,175,710; 8,224,450; and 8,364,278; and U.S. Patent Application Publication No. 2007/0150036, as well as the other references cited above, all of which are incorporated herein by reference.

(30) Electrodes of the lead(s) (or electrodes of a sensor or other device) can be used to sense local electrical characteristics of the environment around the lead(s) and electrodes during and between electrical pulses or waveforms (which can be, for example, therapeutic stimulation pulses or waveforms, sub-perception pulses or waveforms, sensing pulses or waveforms, or other electrical pulses or waveforms). Examples of such sensing can include the measurement, recording, or observation of an electrophysiological signal, such as an evoked potential (EP), evoked compound action potential (ECAP), evoked resonant neural activity (ERNA), local field potential (LFP), ESG (electrospinogram), EEG (electroencephalogram), ECG (electrocardiogram), ECoG (electrocorticogram), or EMG (electromyogram) signal or the like or any combination thereof.

(31) Neural response signals or other neural signals, including, for example, an evoked potential (EP); evoked compound action potential (ECAP); evoked resonant neural activity (ERNA); local field potential (LFP); oscillations within the EEG, ECoG, LFP, or other neural signal; or the like or any combination thereof, can reflect the summed activity of different neural elements. For example, evoked potentials can capture those neurons directly activated by electrical stimulation. Evoked potentials may also reflect the propagation of neural activity across the neural network and may reflect both local and network level activation. FIG. 2A illustrates one example of a neural response **235** or neural signal. When a neural fiber is recruited by electrical stimulation, the neural fiber, in response, will issue an action potential—that is, the neural fiber will “fire.” Should recruitment from electrical stimulation result in the neural fiber's resting state the neural fiber will depolarize, repolarize, and hyperpolarize before coming to rest again. If electrical stimulation continues, the neural fiber will fire again at some later time. FIG. 2B illustrates one example of an evoked compound action potential (ECAP) which can be a cumulative response of neural fibers recruited and firing within a volume.

(32) Although features of the evoked potentials or other neural response signals can provide information about the neural environment, such features typically do not reveal the type of neural fibers that are activated. As described herein, in at least some embodiments, signal decomposition

can be used to, for example, determine or estimate the underlying neural populations activated by stimulation. Use of signal decomposition may assist in improving stimulation targeting procedures. (33) FIG. 3 illustrates one embodiment of a method for estimating neural activation arising from stimulation by a stimulation system. Such estimations may allow for estimation of which neural elements are activated or the amount of activation within a neural element population. In step 302, a user or a system specifies the types or characteristics of neural elements (for example, neural fibers, neural cells, axons, neural terminals, or other neural structures) that may be stimulated. For example, for deep brain stimulation, the specified neural elements may include anatomical structures such as the subthalamic nucleus (STN), globus pallidus interna (GPi), elements of the basal ganglia-thalamo-cortical loop (BGTC), or the like; specific neural tracts; or specific neural cells or cell groups; or the like or any combination thereof. For spinal cord stimulation, the specified neural elements may include, for example, A-beta or A-delta fibers or other specific neural cells or cell groups or neural structures or the like or any combination thereof. As another example, the different neural elements may be defined by one or more characteristics, such as, for example, the diameter, trajectory, orientation, or the like (or any combination thereof) of the neural fibers or neural cells. In at least some embodiments, the user or system designates the number of distinct neural signals that are expected in a neural response signal.

(34) In at least some embodiments, the user can select some or all of the individual neural elements from a menu or a visual display. In at least some embodiments, the user can indirectly specify some or all of the neural elements by selecting anatomical regions, such as regions in the brain. The system can then supply one or more specific neural elements based on the user selection of the anatomical region. In at least some embodiments, knowledge about cells, axons, terminals, or other neural structures around the lead can be used to automatically specify some or all of the neural elements.

(35) FIG. 6 illustrates one embodiment of a user interface 660 for selection of anatomical regions 662 or neural elements. In the illustrated embodiment, the STN anatomical region 662' has been selected. In the illustrated embodiment, the thalamus, GPi, globus pallidus externa (GPe), and internal capsule (IC) are also displayed as well as a representation of a lead 612. In at least some embodiments, the user interface 660 also presents a menu 664 for selection of different types of neural signals to allow the user to select the associated neural signal based on the selected anatomical region or neural element (e.g., the STN. In the illustrated embodiment, the menu 664 includes an Evoked Potential (EP), ERNA, LFP Template 1, and LFP Template 2. Other selections may be presented on the menu 664 depending on the selected anatomical region or neural element and on available templates or signals for the anatomical region or neural element. In at least some embodiments, the user or system can select one or more of these neural signals for the neural element. In at least some embodiments, the anatomical regions may be patient-specific based, for example, on patient images. In at least some embodiments, the anatomical regions may be based on an anatomical atlas or generalized model of the brain. In at least some embodiments, the anatomical regions can be a combination of patient-specific anatomical regions and anatomical regions based on an anatomical atlas or generalized model of the brain.

(36) In step 304, a neural response signal is then recorded or otherwise obtained or measured. In step 306, the recorded neural response signal is decomposed into constituent neural signals for the individual neural elements to identify or determine the neural populations activated by stimulation. Examples of methods for decomposing the recorded neural response signal are described below.

(37) In at least some embodiments, the decomposition of the recorded neural response signal includes assumptions that each of the types of neural elements produce a distinctive signal and that these signals from the neural elements can arrive at different periods in time depending, for example, on the distance from the stimulation source, the threshold amount of stimulation needed to produce a response, distance from the recording arrangement or the like or any combination thereof. As an example, in at least some embodiments, each type of neural element can contribute

distinctive frequency components at specific timepoints in the signal to generate a composite neural response signal.

(38) FIG. 4 illustrates one embodiment of a method for decomposing the recorded neural response signal in the frequency domain. In step 402, the recorded neural response signal is transformed from the time domain to the frequency domain using any suitable technique, such as, for example, any suitable Fourier transform technique (e.g., a short-time Fourier transform (STFT)), or the like or any combination thereof.

(39) In step 404, one or more neural signals (e.g., template signals) for each of the specified types of neural elements are selected, obtained, or otherwise provided. In at least some embodiments, the neural signals for each neural element are signals that have been previously recorded or extracted for those neural elements or are estimated signals generated from simulations, calculations, or models or any combination thereof.

(40) FIG. 7A illustrates one embodiment of a user interface for selecting which neural signals to use for the neural elements. In section 772, the individual neural signals can be selected. In at least some embodiments, the selection can be made from a menu or using up/down arrows 774. In the illustrated embodiment, the STN ERNA, STN EP, and IC EP signals are selected. In optional section 776, the source for a neural signal can be selected by a user including a pre-loaded (e.g., default or stored) signal, an acquired signal, or a custom (e.g., user-provided) signal. In at least some embodiments, if the custom signal is selected, a prompt appears asking the user to upload the custom signal in an appropriate format (e.g., a .dat format). In at least some embodiments, the user or system may select a portion of the neural signal to use in the composite signal. FIG. 7B illustrates examples of neural signals 775 that can be used as templates for the selected neural signals.

(41) In at least some embodiments, the system may send a warning to a user if two or more of the selected neural signals are of different time lengths or are too similar (which may indicate, for example, that the same neural signal has been selected twice).

(42) Returning to step 404 of FIG. 4, the selected neural signals are transformed (if not already transformed) into the frequency domain. These neural signals can be combined into a composite neural signal using individual weights for each of the neural elements to reflect the strength of the signal expected for the population of the specific neural elements. The combined neural signals form an estimated neural signal. The weights can be relative weights (for example, relative to a weight of 1 for one of the neural signals), absolute weights, or weights that sum to 100 (e.g., 100 percent) or 1 or any other suitable number. The combination, and associated weighting, can utilize linear or non-linear models or combinations.

(43) In at least some embodiments, the distinct frequencies for the neural signal(s) from each neural element may be used to create an a priori estimate of the initial weight for each neural element or neural signal. In at least some embodiments, the user or the system may provide an initial estimate of the weight for each of the neural elements. As an example, in the user interface of FIG. 7A, in section 778 of the user interface the user can select the initial relative weights for each of the signals. In at least some embodiments, the initial estimate of the weights can be based on, for example, literature sources, clinician experience, any other suitable source, or the like or any combination thereof. In at least some embodiments, the system may perform an initial weight estimate using any suitable technique including, but not limited to, least squares fitting, generalized linear model (GLM) regression, other regression techniques, QR factorization, machine learning techniques (for example, fuzzy logic, neural networks, k-nearest neighbor (KNN) techniques, density-based unsupervised clustering, Gaussian mixture models, support vector machine-based supervised methods, hierarchical clustering, or the like), coordinate descent, gradient descent, or the like or any combination thereof.

(44) In at least some embodiments, the system may alert a user if neural signals from two or more of the neural elements have known covariance that exceeds or meets a threshold amount. In other

embodiments, the system may alter or select weights based on the known covariance.

(45) FIG. 7D illustrates another embodiment of a user interface **779** that may be provided in addition to, or as an alternative to, section **778** of FIG. 7A. The user interface **779** of FIG. 7D allow the user to select an initial weight for the neural signals that is equal, preset/unequal, or preset/custom. The “preset, unequal” selection in FIG. 7D may be useful when it is known that activation of a certain neural element will intrinsically produce a larger signal than activation of another neural element (for example, due to more of the first neural element than the second or a larger intrinsic response.) If the “preset, custom” selection in FIG. 7D is made, the user can be prompted to enter weights for one or more of the neural signals. In at least some embodiments, if a weight is not entered for the neural signal, a default value (for example, 0, 1, or 100) is used.

(46) In at least some embodiments, the system can filter the neural response signal or the neural signal(s) for each neural element or any combination thereof. In at least some embodiments, the same filters is applied to both the neural response signal and the individual neural signals. The filtering can be based on, for example, lead impedance, device measurement settings (for example, sampling rate or sample averaging), or the like or any combination thereof. In at least some embodiments, the user can set the filter(s). The filters can include, for example, hardware/analog filters, amplifier gain level, artifact blanking settings, least mean square (LMS) artifact cancellation filters, adaptive filters, or digital FIR filters, or the like or any combination thereof.

(47) FIGS. **8** and **9** illustrate filtering or other signal processing of the recorded response signal or the neural signal(s) for the individual neural elements. In FIG. **8**, the acquired response or selected neural signal **880** is illustrated along with a filtered signal **882** obtained by filtering the signal **880**.

(48) FIG. **9** illustrates one embodiment of a user interface **990** for selecting filters or other signal processing. Examples of filters and other signal processing include, but are not limited to, an artifact filter, a noise filter, lead encapsulation/environmental scaling, and template signal properties. The artifact filter can be used, for example, to remove any artifact arising from the stimulation signal that generates the neural response. The artifact filter may include user input such as the pulse width, amplitude, type of stimulation (e.g., biphasic, monophasic, or the like), and the type of filter. The noise filter can be a low pass, high pass, or bandpass filter and may include a user- or system-selected high pass frequency or a user- or system-selected low pass frequency, as appropriate for the type of filter. The lead encapsulation/environmental scaling may account for tissue differences near the lead, such as an encapsulation layer adjacent to the lead, and may provide fixed or frequency-dependent scaling.

(49) The recorded neural response signal may also be filtered based on the properties of the other neural signals such as, for example, sampling rate, sampling duration, or the like or any combination thereof. It will be understood that the filtering or other signal processing can be applied to the recorded neural response signal or the other neural signals or any combination thereof. In at least some embodiments, the same filtering or other signal processing may be applied to both the recorded neural response signal and the other neural signals.

(50) In at least some embodiments, the user may input the lead implant site or estimated distance(s) between the lead and the neural elements. In at least some embodiments, the system may estimate the distance between the lead and the neural elements based on the lead implant site. The system may use the distance information to modify the expected amplitude or other features of any of the neural signal(s) from the individual neural elements.

(51) FIG. **10** illustrates an example of a user interface **1095** with illustrates a lead **1012** and anatomical structures **1062**. The user interface also includes a stimulation field model (SFM) that is calculated based on stimulation parameters (for example, electrode selection, amplitude, pulse width, pulse duration, or the like or any combination thereof) and illustrates the estimated volume of activation (VOA) of tissue that will be activated by the stimulation. Examples of the SFM or VOA, as well as methods and systems for determining the SFM or VOA, can be found at U.S. Pat. Nos. 8,326,433; 8,379,952; 8,649,845; 8,675,945; 8,831,731; 8,849,632; 8,855,773; 8,913,804;

8,918,183; 8,958,615; 9,026,317; 9,050,470; 9,072,905; 9,081,488; 9,084,896; 9,135,400; 9,227,074; 9,235,685; 9,254,387; 9,272,153; 9,302,110; 9,308,372; 9,310,985; 9,364,665; 9,526,902; 9,586,053; 9,792,412; 9,821,167; 9,925,382; and 9,959,940; U.S. Patent Application Publications Nos. 2009/0287272; 2009/0287273; 2012/0314924; 2013/0116744; 2014/0122379; 2015/0066111; 2016/0375248; 2016/0375258; and 2017/0304633; U.S. patent application Ser. Nos. 15/706,004; 15/864,876; and Ser. No. 15/937,264; and U.S. Provisional Patent Application Ser. Nos. 62/030,655 and 62/532,869, all of which are incorporated herein by reference in their entireties.

(52) In at least some embodiments, the user or system may provide or input stimulation settings such as electrode selection, stimulation rate, stimulation duration, pulse width, amplitude, or the like or any combination thereof. The system may use the stimulation settings to adjust the neural signals that arise in response to the stimulation, for example, if those neural signals are known to be sensitive or dependent on one or more of those stimulation settings. FIG. **11** illustrates one embodiment of a user interface **1196** for inputting stimulation parameters such as amplitude, pulse width (PW), or pulse rate.

(53) In step **406**, the difference between the recorded neural response signal and the estimated neural signal is determined. The difference can be determined using any suitable technique such as, for example, mean squared error, a binned mean squared error, mean rectified error, statistical analyze of features of the signal (for example, comparing a mean and standard deviation of a signal feature from template signals versus the mean and standard deviation for the feature of the recorded signal, or the like or any combination thereof. In at least some embodiments, the feasibility, fitness, or robustness of the determined weights can be compared versus a predictor model that can be based on expected physiological models or the like. If the determined weights are not within a threshold difference from the predictor model, the system may alert the user.

(54) In step **408**, the system determines whether the difference is acceptable or not. For example, the system may determine whether the difference meets or exceeds a threshold amount and, if so, whether the difference is unacceptable. If the difference is acceptable, the process ends. FIG. **12** illustrates one example of determined weights for a stimulation: 60% STN, 30% Thalamus, and 10% IC.

(55) If the difference is not acceptable, the weights are adjusted in step **410** and the process returns to step **404**. In at least some embodiments, the weight adjustment may be requested from the user. In at least some embodiments, the system may perform the weight adjustment using any suitable technique including, but not limited to, least squares fitting, generalize linear model (GLM) regression, other regression techniques, QR factorization, machine learning techniques (for example, fuzzy logic, neural networks, k-nearest neighbor (KNN) techniques, density-based unsupervised clustering, Gaussian mixture models, support vector machine-based supervised methods, hierarchical clustering, or the like or any combination thereof), coordinate descent, gradient descent, or the like or any combination thereof. The process can continue until acceptable weights are found or the system or user halts the process.

(56) FIG. **5** illustrates one embodiment of a method for decomposing the recorded signal in the time domain. In step **502**, template signals for each of the specified types of neural elements are selected. FIG. **7B** illustrates examples of neural signals **775** that can be used as templates for the selected neural signals. In at least some embodiments, the template signals for each neural element are signals that have been previously recorded or extracted for those neural elements or are estimated signals generated from simulations, calculations, or models or any combination thereof. The same considerations and options described above with respect to the method illustrated in FIG. **4** are applicable to the method illustrated in FIG. **5**, including all of the considerations and options illustrated in FIG. **7A** to FIG. **12**.

(57) In step **504**, weights for the template signals are determined from the recorded neural signal. The system performs a weight determination using any suitable technique including, but not

limited to, least squares fitting, generalized linear model (GLM) regression, other regression techniques, QR factorization, machine learning techniques (for example, fuzzy logic, neural networks, k-nearest neighbor (KNN) techniques, density-based unsupervised clustering, Gaussian mixture models, support vector machine-based supervised methods, hierarchical clustering, or the like), coordinate descent, gradient descent, or the like or any combination thereof. FIG. 7C illustrates a composite signal 777 of the three template signals of FIG. 7B combined using weights (for example, 10% of S.sub.1, 30% of S.sub.2, and 60% of S.sub.3), as described in more detail below.

(58) In at least some embodiments, the process ends with step 504. In other embodiments, the resulting composite signal may be compared with the neural response signal or a predicted or estimated neural signal. In at least some other embodiments, the user or the system may provide a predicted estimate of the weight for each of the neural elements using, for example, literature sources, clinician experience, any other suitable source, or the like or any combination thereof. In at least some embodiments, the feasibility, fitness, or robustness of the determined weights can be compared versus a predictor model that can be based on expected physiological models or the like. (59) In optional step 506, the difference between the recorded neural response signal and composite signal generated using the estimated or predicted weights (or the predicted or estimated neural signal) is determined. The difference can be determined using any suitable technique such as, for example, mean squared error, a binned mean squared error, mean rectified error, statistical analyze of features of the signal (for example, comparing a mean and standard deviation of a signal feature from template signals versus the mean and standard deviation for the feature of the recorded signal, or the like or any combination thereof).

(60) In optional step 508, the system determines whether the difference is acceptable or not. For example, the system may determine whether the difference meets or exceeds a threshold amount and, if so, the difference is unacceptable. For example, in at least some embodiments, if the determined weights are not within a threshold difference from the predictor model, the system may alert the user. If the difference is acceptable, the process ends.

(61) If the difference is not acceptable, the weights are adjusted or recalculated in optional step 510 and the process returns to step 506. In at least some embodiments, the weight adjustment may be requested from the user. In at least some embodiments, the system may perform the weight adjustment using any suitable technique including, but not limited to, least squares fitting, generalize linear model (GLM) regression, other regression techniques, QR factorization, machine learning techniques (for example, fuzzy logic, neural networks, k-nearest neighbor (KNN) techniques, density-based unsupervised clustering, Gaussian mixture models, support vector machine-based supervised methods, hierarchical clustering, or the like), coordinate descent, gradient descent, or the like or any combination thereof. The process can continue until acceptable weights are found or the system or user halts the process.

(62) It will be understood that the methods illustrated in FIGS. 4 and 5 can be modified to apply to either time domain or frequency domain methods or any combination thereof. It will be understood that each block of the flowchart illustration, and combinations of blocks in the flowchart illustration and methods disclosed herein, can be implemented by computer program instructions. These program instructions may be provided to a processor to produce a machine, such that the instructions, which execute on the processor, create means for implementing the actions specified in the flowchart block or blocks disclosed herein. The computer program instructions may be executed by a processor to cause a series of operational steps to be performed by the processor to produce a computer implemented process. The computer program instructions may also cause at least some of the operational steps to be performed in parallel. Moreover, some of the steps may also be performed across more than one processor, such as might arise in a multi-processor computing device. In addition, one or more processes may also be performed concurrently with other processes, or even in a different sequence than illustrated without departing from the scope or

spirit of the invention.

(63) The computer program instructions can be stored on any suitable computer-readable medium including, but not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (“DVD”) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, cloud-based storage, or any other medium which can be used to store the desired information and which can be accessed by a computing device.

(64) FIG. 13 is a schematic overview of one embodiment of components of an electrical stimulation system **1300** including an electronic subassembly **1358** disposed within a control module. The electronic subassembly **1358** may include one or more components of the IPG. It will be understood that the electrical stimulation system can include more, fewer, or different components and can have a variety of different configurations including those configurations disclosed in the stimulator references cited herein.

(65) Some of the components (for example, a power source **1312**, an antenna **1318**, a receiver **1302**, and a processor **1304**) of the electrical stimulation system can be positioned on one or more circuit boards or similar carriers within a sealed housing of an implantable pulse generator (see e.g., **14** in FIG. 1), if desired. Any power source **1312** can be used including, for example, a battery such as a primary battery or a rechargeable battery. Examples of other power sources include super capacitors, nuclear or atomic batteries, mechanical resonators, infrared collectors, thermally-powered energy sources, flexural powered energy sources, bioenergy power sources, fuel cells, bioelectric cells, osmotic pressure pumps, and the like including the power sources described in U.S. Pat. No. 7,437,193, incorporated herein by reference.

(66) As another alternative, power can be supplied by an external power source through inductive coupling via the optional antenna **1318** or a secondary antenna. In at least some embodiments, the antenna **1318** (or the secondary antenna) is implemented using the auxiliary electrically-conductive conductor. The external power source can be in a device that is mounted on the skin of the user or in a unit that is provided near the user on a permanent or periodic basis.

(67) If the power source **1312** is a rechargeable battery, the battery may be recharged using the optional antenna **1318**, if desired. Power can be provided to the battery for recharging by inductively coupling the battery through the antenna to a recharging unit **1316** external to the user. Examples of such arrangements can be found in the references identified above. The electronic subassembly **1358** and, optionally, the power source **1312** can be disposed within a control module (e.g., the IPG **14** or the ETS **20** of FIG. 1).

(68) In one embodiment, electrical stimulation signals are emitted by the electrodes **26** to stimulate nerve fibers, muscle fibers, or other body tissues near the electrical stimulation system. The processor **1304** is generally included to control the timing and electrical characteristics of the electrical stimulation system. For example, the processor **1304** can, if desired, control one or more of the timing, frequency, strength, duration, and waveform of the pulses. In addition, the processor **1304** can select which electrodes can be used to provide stimulation, if desired. In some embodiments, the processor **1304** selects which electrode(s) are cathodes and which electrode(s) are anodes. In some embodiments, the processor **1304** is used to identify which electrodes provide the most useful stimulation of the desired tissue.

(69) Any processor can be used and can be as simple as an electronic device that, for example, produces pulses at a regular interval or the processor can be capable of receiving and interpreting instructions from an external programming unit **1308** that, for example, allows modification of pulse characteristics. In the illustrated embodiment, the processor **1304** is coupled to a receiver **1302** which, in turn, is coupled to the optional antenna **1318**. This allows the processor **1304** to receive instructions from an external source to, for example, direct the pulse characteristics and the selection of electrodes, if desired.

(70) In one embodiment, the antenna **1318** is capable of receiving signals (e.g., RF signals) from an

external telemetry unit **1306** which is programmed by the programming unit **1308**. The programming unit **1308** can be external to, or part of, the telemetry unit **1306**. The telemetry unit **1306** can be a device that is worn on the skin of the user or can be carried by the user and can have a form similar to a pager, cellular phone, or remote control, if desired. As another alternative, the telemetry unit **1306** may not be worn or carried by the user but may only be available at a home station or at a clinician's office. The programming unit **1308** can be any unit that can provide information to the telemetry unit **1306** for transmission to the electrical stimulation system **1300**. The programming unit **1308** can be part of the telemetry unit **1306** or can provide signals or information to the telemetry unit **1306** via a wireless or wired connection. One example of a suitable programming unit is a computer operated by the user or clinician to send signals to the telemetry unit **1306**.

(71) The signals sent to the processor **1304** via the antenna **1318** and the receiver **1302** can be used to modify or otherwise direct the operation of the electrical stimulation system. For example, the signals may be used to modify the pulses of the electrical stimulation system such as modifying one or more of pulse duration, pulse frequency, pulse waveform, and pulse strength. The signals may also direct the electrical stimulation system **1300** to cease operation, to start operation, to start charging the battery, or to stop charging the battery. In other embodiments, the stimulation system does not include the antenna **1318** or receiver **1302** and the processor **1304** operates as programmed.

(72) Optionally, the electrical stimulation system **1300** may include a transmitter (not shown) coupled to the processor **1304** and the antenna **1318** for transmitting signals back to the telemetry unit **1306** or another unit capable of receiving the signals. For example, the electrical stimulation system **1300** may transmit signals indicating whether the electrical stimulation system **1300** is operating properly or not or indicating when the battery needs to be charged or the level of charge remaining in the battery. The processor **1304** may also be capable of transmitting information about the pulse characteristics so that a user or clinician can determine or verify the characteristics.

(73) FIG. **14** illustrates one embodiment of a system for practicing any of the methods described herein. The system can include a computer **1400** or any other similar device that includes at least one processor **1402** and a memory **1404**, a display **1406**, an input device **1408**, and, optionally, the electrical stimulation system **1412**.

(74) The computer **1400** can be a laptop computer, desktop computer, tablet, mobile device, smartphone, or other devices that can run applications or programs, or any other suitable device for processing information and for presenting a user interface. The computer can be, for example, the CP **18** or RC **16** of FIG. **1** or the programming unit **1308** of FIG. **13**. The computer **1400** can be local to the user or can include components that are non-local to the user including one or both of the processor **1402** or memory **1404** (or portions thereof). For example, in some embodiments, the user may operate a terminal that is connected to a non-local computer. In other embodiments, the memory can be non-local to the user. As another example, the computer **1400** may utilize or communicate with a processor in the control module **1414** (such as the IPG **14** or ETS **20** of FIG. **1**).

(75) The computer **1400** can utilize any suitable processor **1402** including one or more hardware processors that may be local to the user or non-local to the user or other components of the computer. The processor **1402** is configured to execute instructions provided to the processor, as described below.

(76) Any suitable memory **1404** can be used for the processor **1402**. The memory **1404** illustrates a type of computer-readable media, namely computer-readable storage media. Computer-readable storage media may include, but is not limited to, nonvolatile, non-transitory, removable, and non-removable media implemented in any method or technology for storage of information, such as computer readable instructions, data structures, program modules, or other data. Examples of computer-readable storage media include RAM, ROM, EEPROM, flash memory, or other memory

technology, CD-ROM, digital versatile disks (“DVD”) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by a computer.

(77) Communication methods provide another type of computer readable media; namely communication media. Communication media typically embodies computer-readable instructions, data structures, program modules, or other data in a modulated data signal such as a carrier wave, data signal, or other transport mechanism and include any information delivery media. The terms “modulated data signal,” and “carrier-wave signal” includes a signal that has one or more of its characteristics set or changed in such a manner as to encode information, instructions, data, and the like, in the signal. By way of example, communication media includes wired media such as twisted pair, coaxial cable, fiber optics, wave guides, and other wired media and wireless media such as acoustic, RF, infrared, and other wireless media.

(78) The display **1406** can be any suitable display device, such as a monitor, screen, display, or the like, and can include a printer. The input device **1408** can be, for example, a keyboard, mouse, touch screen, track ball, joystick, voice recognition system, or any combination thereof, or the like and can be used by the user to interact with a user interface or clinical effects map.

(79) The electrical stimulation system **1412** can include, for example, a control module **1414** (for example, the IPG **14** or ETS **20** of FIG. **1**) and a lead **1416** (for example, the lead **12** illustrated in FIG. **1**.) The electrical stimulation system **1412** may communicate with the computer **1400** through a wired or wireless connection or, alternatively or additionally, a user can provide information between the electrical stimulation system **1412** and the computer **1400** using a computer-readable medium or by some other mechanism. In some embodiments, the computer **1400** may include part of the electrical stimulation system. In at least some embodiments, the computer **1400** can program the control module **1414** for delivery of stimulation pulses, charge recovery pulse, charge recovery phases, or the like or any combination thereof.

(80) The above specification and examples provide a description of the manufacture and use of the invention. Since many embodiments of the invention can be made without departing from the spirit and scope of the invention, the invention also resides in the claims hereinafter appended.

Claims

1. A method for estimating neural activation arising from deep brain stimulation by a stimulation system, the method comprising: identifying a plurality of different anatomical structures of a brain stimulated by the deep brain stimulation; for each of the different anatomical structures of the brain, providing at least one template signal for the anatomical structure of the brain; obtaining a neural response signal resulting from the deep brain stimulation by the stimulation system; decomposing the neural response signal in a time domain into a plurality of decomposed signals using the template signals to estimate neural activation of each of the different anatomical structures of the brain, wherein decomposing the neural response signal comprises decomposing the neural response signal using the template signals, wherein decomposing the neural response signal comprises determining weights for each of the template signals as portions of the neural response signal; combining the template signals using the weights to generate a composite neural signal; and using the estimated neural activations of each of the different anatomical structures of the brain to adjust stimulation settings of the stimulation system.
2. The method of claim 1, further comprising determining a difference between the neural response signal and the composite neural signal.
3. The method of claim 2, further comprising determining when the difference exceeds a predetermined threshold difference and, when so, adjusting the weights.
4. The method of claim 1, further comprising filtering the neural response signal.
5. The method of claim 1, wherein the plurality of different anatomical structures of the brain

comprises at least one of the thalamus, subthalamic nucleus, globus pallidus interna, globus pallidus externa, or internal capsule.

6. The method of claim 1, further comprising displaying a representation of at least a portion of the brain and receiving selections by a user from the representation of the plurality of different anatomical structures of the brain for deep brain stimulation.

7. The method of claim 1, further comprising, for each of the different anatomical structures of the brain, receiving a selection of a type of signal to be used for the template signal.

8. A method for estimating neural activation arising from deep brain stimulation by a stimulation system, the method comprising: identifying a plurality of different anatomical structures of a brain stimulated by the deep brain stimulation; for each of the different anatomical structures of the brain, providing at least one template signal for the anatomical structure of the brain; determining initial weights for each of the template signals using one or more distinctive frequency components in the template signals; obtaining a neural response signal resulting from the deep brain stimulation by the stimulation system; transforming the neural response signal into a frequency domain; decomposing the neural response signal in the frequency domain into a plurality of decomposed signals using the template signals to estimate neural activation of each of the different anatomical structures of the brain, wherein decomposing the neural response signal comprises determining weights for each of the template signals as portions of the neural response signal; and using the estimated neural activations of each of the different anatomical structures of the brain to adjust stimulation settings of the stimulation system.

9. The method of claim 8, further comprising displaying a representation of at least a portion of the brain and receiving selections by a user from the representation of the plurality of different anatomical structures of the brain for deep brain stimulation.

10. The method of claim 8, further comprising, for each of the different anatomical structures of the brain, receiving a selection of a type of signal to be used for the template signal.

11. The method of claim 8, further comprising filtering the neural response signal.

12. The method of claim 8, wherein the plurality of different anatomical structures of the brain comprises at least one of the thalamus, subthalamic nucleus, globus pallidus interna, globus pallidus externa, or internal capsule.

13. The method of claim 8, further comprising combining the template signals using the weights to generate a composite neural signal.

14. A method for estimating neural activation arising from deep brain stimulation by a stimulation system, the method comprising: identifying a plurality of different anatomical structures of a brain stimulated by the deep brain stimulation; for each of the different anatomical structures of the brain, providing at least one template signal for the anatomical structure of the brain; obtaining a neural response signal resulting from the deep brain stimulation by the stimulation system; transforming the neural response signal into the frequency domain; decomposing the neural response signal in the frequency domain into a plurality of decomposed signals using the template signals to estimate neural activation of each of the different anatomical structures of the brain, wherein decomposing the neural response signal comprises determining weights for each of the template signals as portions of the neural response signal; combining the template signals using the weights to generate a composite neural signal; and using the estimated neural activations of each of the different anatomical structures of the brain to adjust stimulation settings of the stimulation system.

15. The method of claim 14, further comprising determining a difference between the neural response signal and the composite neural signal.

16. The method of claim 15, further comprising determining when the difference exceeds a predetermined threshold difference and, when so, adjusting the weights.

17. The method of claim 14, further comprising displaying a representation of at least a portion of the brain and receiving selections by a user from the representation of the plurality of different

anatomical structures of the brain for deep brain stimulation.

18. The method of claim 14, further comprising, for each of the different anatomical structures of the brain, receiving a selection of a type of signal to be used for the template signal.

19. The method of claim 14, further comprising filtering the neural response signal.

20. The method of claim 14, wherein the plurality of different anatomical structures of the brain comprises at least one of the thalamus, subthalamic nucleus, globus pallidus interna, globus pallidus externa, or internal capsule.
